Shaowei Zhang

shaowei.zhang@stud.uni-heidelberg.de · (+49)17656343362

Education

Heidelberg University *M.A. in Computational Linguistics* 2024 – Expected Jun. 2027

Selected Coursework: Algorithmic Methods in NLP, Machine Learning, Linguistic Representations, Language Learning Systems

Xidian University *B.Eng. in Artificial Intelligence* 2020 – 2024

Selected Coursework: Optimization, Pattern Recognition, Deep Learning, Knowledge Engineering, Natural Language Processing, Database Systems
Bachelor Thesis: Loss Function Learning for Classification Tasks

Internship Experience

OneFlow Inc. & SiliconFlow Inc. Feb. 2024 – Aug. 2024

R&D Engineer Intern (Distributed Training / Systems)

- Contributed to the development of an internal fine-tuning framework built on Libai / OneFlow, extending large-scale training workflows with supervised fine-tuning and parameter-efficient adaptation methods.
- Supported fine-tuning workflows for multiple model families, covering both full-parameter and PEFT-based adaptation, while improving maintainability through refactoring and more consistent configuration handling.
- Investigated multi-GPU scaling behavior in OneFlow + Libai and participated in framework migration to heterogeneous platforms.

Beijing PERCENT Technology Group Co., Ltd. Jan. 2023 – Dec. 2023

Machine Learning Intern (NLP / LLM)

- Worked on both machine translation and LLM-related development as the team shifted toward large language model applications and adaptation workflows.
- Built and maintained training and evaluation pipelines for machine translation and language models, covering data processing, fine-tuning, continued pretraining, and downstream assessment.
- Conducted experiments on parameter-efficient adaptation and model evaluation, with additional exposure to practical inference tooling such as vLLM and TensorRT.

Skills

- **Programming:** Python, C++; familiar with Java and C
- **ML / Frameworks:** PyTorch, OneFlow; fine-tuning, PEFT, continued pretraining, evaluation workflows
- **Systems:** Linux/Unix, Git, Slurm, Distributed training fundamentals: debugging, profiling
- **Languages:** Mandarin, English

Research Interests

- Efficient machine learning systems, especially large-scale training, fine-tuning, and inference workflows
- Parameter-efficient adaptation methods for language models
- Reliable NLP / LLM systems, with an emphasis on evaluation, robustness, and reproducible research prototyping